

Beyond Packet Error Rate: A Systematic Analysis of Performance Evaluation and KPI Coverage in DECT-2020 NR Research

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Abstract—DECT-2020 New Radio (DECT-2020 NR) is a non-cellular International Mobile Telecommunications-2020 (IMT-2020) radio technology supporting decentralized and infrastructure-light wireless communication. Although research has progressed from analytical and simulation-based studies toward practical implementations, its experimental maturity has not been systematically assessed.

This paper analyzes the experimental maturity of DECT-2020 NR based on 51 primary studies published between 2020 and July 2026. Simulation remains the dominant evaluation methodology, appearing in 32 studies, while 16 studies employ experimental or testbed-based evaluation. Hardware prototypes are reported in 16 studies and commercial-device evaluations in 11, whereas only five studies report field trials and three report industrial measurements. A cross-analysis of technical areas and validation approaches reveals uneven maturity: Physical Layer (PHY) research shows comparatively strong hardware and practical validation, while routing and mesh networking remain predominantly simulation-based and scheduling and resource allocation have limited experimental evidence. Based on these findings, a five-level experimental-maturity framework is proposed, and large-scale physical testbeds, application-oriented validation, long-duration measurements, and reproducible benchmarking are identified as key priorities for future DECT-2020 NR research.

I. INTRODUCTION

Next-generation wireless networks increasingly support Internet of Things (IoT) and industrial applications with heterogeneous requirements for latency, reliability, energy efficiency, scalability, coverage, mobility, and predictability [1], [2]. Within this landscape, DECT-2020 NR has emerged as a non-cellular IMT-2020 radio technology

designed for decentralized and infrastructure-light wireless communication. Its flexible device roles, autonomous radio-resource operation, and support for direct and multi-hop communication make it particularly relevant to IoT and Industrial Internet of Things (IIoT) deployments requiring local network ownership, dense connectivity, and flexible network topologies [3]–[5].

Since its introduction, research on DECT-2020 NR has expanded substantially. Early work concentrated primarily on IMT-2020 candidate-technology assessment, link-level performance, and compliance with communication requirements [6]–[8]. Subsequent research has addressed a broader range of topics, including PHY and Medium Access Control (MAC) mechanisms, multi-hop and mesh networking, routing, resource management, reliability, energy efficiency, positioning, coexistence, and application-oriented implementations. At the same time, evaluation has begun to progress beyond analytical and simulation-based investigation toward Software-Defined Radio (SDR) and Field-Programmable Gate Array (FPGA) implementations, commercial devices, hardware prototypes, field trials, and industrial measurements [9]–[12].

This progression from theoretical investigation toward physical validation is particularly important for assessing the practical capabilities of DECT-2020 NR. Analytical models and simulation provide essential means for investigating protocol behavior, scalability, and performance under controlled and reproducible assumptions. However, they cannot fully capture implementation- and deployment-dependent effects such as transceiver turnaround, processing overhead, synchronization behavior, practical propagation and interference, hardware energy consumption, radio-state transitions, and

implementation-specific protocol interactions. These effects become increasingly important when evaluating multi-device, multi-hop, and latency- or reliability-sensitive communication under realistic operating conditions.

Despite the increasing number of experimental studies, the extent to which DECT-2020 NR research has progressed toward physical, integrated-system, and real-world validation has not been systematically characterized across technical areas. Existing work provides evidence for individual mechanisms and use cases, but does not reveal whether experimental maturity has developed uniformly across the research landscape. Consequently, it remains unclear which technical areas are supported by substantial hardware and real-world evidence, which remain predominantly dependent on analytical or simulation-based evaluation, and where the principal experimental validation gaps persist.

To address this gap, this paper performs a focused secondary analysis of a systematically established corpus of 51 DECT-2020 NR studies published between 2020 and July 2026. Rather than providing another general technical survey, the study examines how the available evidence progresses from analytical feasibility and simulation toward prototype, integrated-system, and industrial validation. The main contributions of this paper are:

- a quantitative mapping of evaluation methodologies and validation approaches across a systematically established corpus of 51 DECT-2020 NR studies;
- a cross-analysis of experimental maturity across major technical research areas, identifying where research volume is supported by physical and real-world evidence and where validation remains predominantly analytical or simulation-based; and
- a five-level experimental-maturity framework relating analytical feasibility, simulation validation, prototype validation, integrated-system validation, and industrial integration, together with research priorities for reproducible, application-oriented, and real-world DECT-2020 NR validation.

The remainder of this paper is organized as follows. Section II describes the systematic mapping methodology. Section III analyzes the evaluation landscape and experimental maturity across technical areas. Section IV introduces the proposed maturity framework, and Section V concludes the paper.

II. SYSTEMATIC MAPPING METHODOLOGY

This study analyzes the experimental maturity of DECT-2020 NR using a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-based Systematic Literature Review (SLR) corpus covering publications from January 2020 to July 2026. The underlying review process was designed to identify, screen, classify, and synthesize research addressing the architecture, protocols, implementation, and performance evaluation of

DECT-2020 NR. Normative European Telecommunications Standards Institute (ETSI) specifications and relevant International Telecommunication Union Radiocommunication Sector (ITU-R) recommendations were analyzed separately to provide technical and standardization context and were not counted as primary research studies in the quantitative analysis.

The experimental-maturity analysis addresses the following research questions:

- RQ1** How has DECT-2020 NR research progressed from analytical and simulation-based evaluation toward experimental and real-world validation?
- RQ2** Which technical areas have accumulated experimental evidence, and which remain predominantly dependent on simulation or analytical evaluation?
- RQ3** Which experimental and methodological gaps must be addressed to enable reproducible and application-oriented validation?

A. Corpus and Study Selection

IEEE Xplore, Scopus, ACM Digital Library, Springer-Link, and ScienceDirect were searched for relevant publications. Google Scholar was used as a complementary source for backward and forward snowballing rather than as an additional primary database. The search covered publications from January 2020 to July 2026 and used the following Boolean query:

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("DECT-2020 NR" OR "DECT NR+" OR "DECT-2020 New Radio" OR "DECT NR" OR "DECT New Radio") AND (architecture OR protocol OR PHY OR "physical layer" OR MAC OR "medium access control" OR implementation OR experimental OR testbed OR prototype OR performance OR evaluation OR measurement OR industrial OR IoT OR URLLC OR reliability)
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Studies were considered eligible when they investigated DECT-2020 NR as a primary technology and reported a technical, methodological, implementation-oriented, or performance-related contribution. Publications not primarily concerned with DECT-2020 NR, duplicate records, and documents without sufficient technical relevance to the research questions were excluded. Normative standards and recommendations were retained separately as contextual sources and were excluded from the primary-study count.

The primary database searches identified 107 records: 42 from IEEE Xplore, 44 from Scopus, one from ACM Digital Library, 15 from SpringerLink, and five from ScienceDirect. An additional 226 records were examined through complementary snowballing. After duplicate handling, title and abstract screening, retrieval, and full-text eligibility assessment, 51 primary studies were retained for the systematic mapping. The complete study-selection process is shown in Fig. 1.

B. Experimental-Maturity Coding

Each of the 51 primary studies was systematically coded along five dimensions: technical topic, evaluation

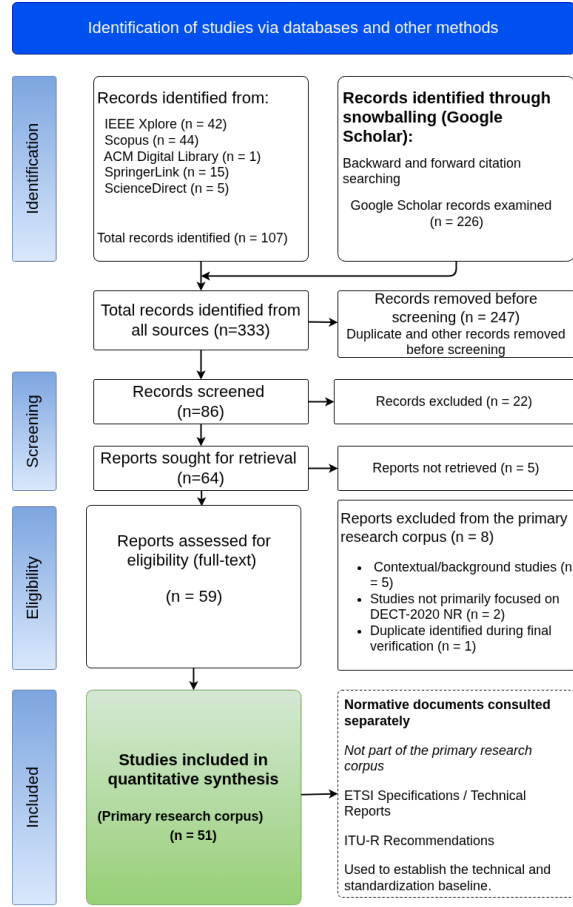


Fig. 1. PRISMA flow diagram of the study-selection process.

methodology, validation approach, implementation platform, and reported Key Performance Indicators (KPIs). The coding scheme was multi-label, allowing a study to contribute to more than one technical area, methodology, validation approach, platform, or performance metric when supported by the reported evaluation.

Technical topics included PHY, MAC, routing and mesh networking, scheduling and resource allocation, energy efficiency, positioning, security and multi-RAT integration, coexistence and regulation, Ultra-Reliable Low-Latency Communications (URLLC), and Massive Machine-Type Communications (mMTC). Evaluation methodologies were coded as simulation, analytical/modeling, experimental/testbed, or survey/overview according to the methods reported by each study.

For the experimental-maturity analysis, validation evidence was grouped into five categories. *Analytical* evidence comprises mathematical and model-based investigation, while *simulation* captures software-based numerical or event-driven evaluation. *Prototype/hardware* evidence combines hardware prototypes and SDR/over-the-air implementations. *Commercial device/testbed* identifies evaluations using commercial devices, and *field/industrial* combines field trials and industrial measurements. These

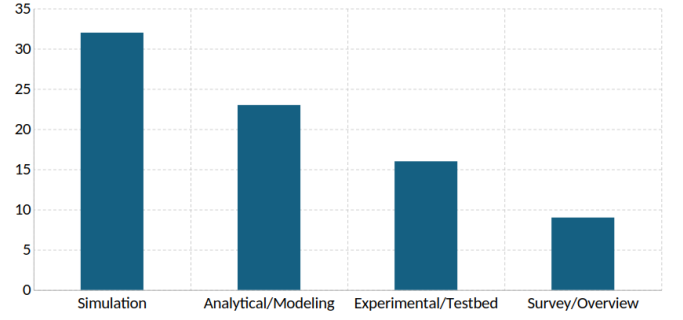


Fig. 2. Evaluation methodologies employed across the 51 included studies (multi-label).

categories were cross-analyzed with the technical topics to determine where research has progressed toward physical and real-world validation.

Because both technical topics and validation approaches are multi-label, category totals are not mutually exclusive and therefore do not sum to the 51 included studies. This coding strategy enables the analysis to distinguish research volume from experimental maturity: a frequently investigated technical area may still have limited prototype, testbed, or field-level evidence.

III. EXPERIMENTAL MATURITY OF DECT-2020 NR RESEARCH

A. Evaluation Landscape

The 51-study corpus shows substantial recent growth, with 38 studies (approximately 75%) published between 2024 and July 2026. Simulation remains the dominant evaluation methodology, appearing in 32 studies, followed by analytical/modeling approaches in 23 studies and experimental/testbed evaluation in 16. Nine studies additionally employ survey or overview methodologies. Because these categories are multi-label, individual studies may employ multiple evaluation approaches.

Figure 2 shows that, despite increasing experimental activity, analytical and simulation-based evaluation still constitutes a substantial part of the evidence base. The presence of experimental/testbed evaluation in 16 studies nevertheless indicates a visible transition from feasibility-oriented investigation toward physical implementation and practical validation, although this transition is uneven across technical research areas.

B. Experimental Maturity Across Technical Areas

Table I cross-analyzes the major technical areas with their validation approaches. Prototype/hardware evidence combines hardware prototypes and SDR/over-the-air validation, while field/industrial evidence combines field trials and industrial measurements. Because both technical areas and validation approaches are multi-label, the values should be interpreted as overlapping evidence rather than mutually exclusive study counts.

TABLE I

EXPERIMENTAL MATURITY ACROSS MAJOR DECT-2020 NR RESEARCH AREAS IN THE 51-STUDY CORPUS. CATEGORIES ARE MULTI-LABEL; THEREFORE, INDIVIDUAL STUDIES MAY CONTRIBUTE TO MULTIPLE RESEARCH AREAS AND VALIDATION APPROACHES.

Research Area	Studies	Analytical	Simulation	Prototype/Hardware	Commercial Device/Testbed	Field/Industrial
PHY	22	10	10	11	8	5
MAC	22	11	16	4	2	0
Routing/Mesh	23	12	17	3	1	1
Scheduling/Resources	4	3	3	1	1	0
Energy	8	2	3	5	4	2
Positioning	6	6	4	2	2	1
Security/Multi-RAT	3	0	0	3	0	0
Coexistence/Regulation	2	0	2	0	0	0
URLLC	9	3	8	1	1	0
mMTC	5	2	5	0	0	0

The results reveal substantial differences between research volume and experimental maturity. Routing/mesh constitutes the largest technical cluster, yet 17 of its 23 studies employ simulation and only three provide prototype/hardware evidence. In contrast, PHY research shows the strongest progression toward physical validation, with 11 of 22 studies reporting prototype/hardware evidence, eight involving commercial-device/testbed validation, and five reaching field/industrial evaluation. MAC, URLLC, and mMTC remain considerably more simulation-oriented.

A notable pattern is that research volume and experimental maturity do not evolve proportionally. Routing/mesh and MAC are among the most frequently investigated areas, yet their evidence remains dominated by simulation. In contrast, energy research has a smaller publication base but a comparatively high proportion of hardware and commercial-device validation. This difference may partly reflect the practical complexity of experimental validation. Device- and link-level behavior can often be investigated using relatively small setups, whereas routing, scheduling, and multi-hop mechanisms require larger synchronized deployments, controlled multi-device traffic, and repeatable network-level experiments. Research volume alone is therefore not a sufficient indicator of practical validation maturity.

C. PHY and MAC

Early PHY research concentrated on candidate-technology assessment, IMT-2020 compliance, link-level behavior, modulation and coding, and system-level performance [6]–[8], [13]–[15]. Subsequent studies extended this evidence toward channel estimation, Non-Orthogonal Multiple Access (NOMA), propagation and range measurements, industrial-channel characterization, FPGA-oriented implementation, and commercial-hardware evaluation [9]–[12], [16]–[21].

The maturity profile of MAC research is less experimental. Although MAC aspects appear in 22 studies, 16 employ simulation and only four provide prototype/hardware evidence. Existing work addresses outage reduction, relaying, random access, coexistence, reliability, and multi-hop access [22]–[27]. Measurement-driven

results further show that hardware effects such as RX–TX turnaround and single-transceiver operation can directly affect UL/DL scheduling latency [28], emphasizing the need for physical multi-device validation.

This distinction is particularly important for MAC evaluation. Simulation can characterize protocol logic, channel access, and resource-allocation behavior under controlled assumptions, but practical performance is additionally influenced by implementation-dependent effects. Processing delays, synchronization, radio-state transitions, channel sensing, and transceiver turnaround can affect application-visible latency and reliability. Physical validation is therefore particularly important when assessing URLLC-oriented or deterministic communication mechanisms whose performance depends on interactions between protocol operation and hardware behavior.

D. Routing, Mesh, and Resource Allocation

Routing and mesh networking constitute the largest technical cluster, with 23 studies addressing flooding, source routing, relay assignment, deep mesh topologies, reliability-oriented path selection, and multi-hop forwarding [4], [29]–[38].

Despite this research volume, 17 studies rely on simulation, while only three provide prototype/hardware evidence and one reaches field/industrial validation. Large-scale physical mesh validation therefore remains a major gap. This is particularly relevant because behavior observed in simulated multi-hop networks may become more difficult to reproduce as physical network depth and device density increase. Contention, accumulated forwarding delay, interference, route instability, synchronization effects, and concurrent traffic may interact across multiple hops. Larger physical topologies are therefore needed to determine whether the scalability demonstrated in simulation translates into predictable end-to-end performance under implementation and deployment constraints.

Scheduling and resource allocation form a much smaller evidence base, appearing in only four studies [21], [25], [38], [39]. Only one provides prototype/hardware and commercial-device evidence. Together with measurement-driven observations of practical transceiver constraints

[28], this indicates a need for application-oriented scheduling evaluated on physical multi-device systems.

The limited experimental evidence in this area is particularly relevant to industrial communication because scheduling and resource allocation connect lower-layer radio capabilities with application-visible performance. Decisions concerning resource assignment, transmission opportunities, retransmissions, and UL/DL operation can directly influence latency, reliability, throughput, and energy consumption. Consequently, evaluating these mechanisms under heterogeneous traffic and multiple simultaneously active devices represents an important step toward application-oriented DECT-2020 NR validation.

E. Application-Oriented and Emerging Areas

Energy-related research exhibits comparatively strong experimental maturity: five of eight studies contain prototype/hardware evidence and four use commercial-device validation. Existing work addresses modem energy consumption, multi-hop energy trade-offs, router rotation, and energy-aware operation [11], [12], [40]–[42]. Positioning combines analytical and simulation-based investigations with a smaller practical evidence base [12], [18], [43], [44].

Multi-RAT integration forms a small but experimentally oriented cluster, particularly through DECT/Wi-Fi integration and unified radio-access implementations [45], [46]. Professional-audio implementations provide additional practical evidence for application-oriented validation [47]. In contrast, coexistence/regulatory studies remain predominantly simulation-oriented [25], [48].

The broader application landscape includes professional audio, tactile communication, backscatter-assisted communication, and LPWAN-oriented comparisons [49]–[54]. These studies demonstrate the breadth of potential DECT-2020 NR applications and deployment scenarios, but the available evidence remains fragmented across application types, hardware platforms, and experimental configurations. Broader application-oriented evaluation is therefore needed to determine how technical mechanisms behave under traffic patterns and operational requirements representative of different industrial and IoT use cases.

F. Experimental Platforms and Performance Evidence

The transition toward physical validation is reflected in the experimental platforms: 16 studies report hardware prototypes, 11 use commercial devices, five report field trials, and three include industrial measurements. Nordic nRF91-series devices are the most frequently identified commercial hardware platform, while other studies employ SDR, Universal Software Radio Peripheral (USRP), GNU Radio, MATLAB, Zephyr, and FPGA-oriented implementations.

Performance evaluation remains concentrated on packet error or loss, reported in 40 studies, followed by

energy/power in 22, latency in 15, and reliability/outage in 15. Throughput appears in only five studies, while application-level indicators such as fairness, Age of Information (AoI), localization error, connection density, and spectral efficiency are considerably less frequent. This distribution indicates that future industrial validation should increasingly consider multiple performance dimensions jointly, including latency, reliability, throughput, jitter, energy consumption, mobility, and network lifetime, rather than relying primarily on isolated average KPIs.

A further challenge is the heterogeneity of experimental setups. Existing studies differ in hardware platforms, network size, topology, traffic generation, radio configuration, propagation environment, software and firmware implementation, and measurement duration. Such diversity demonstrates the breadth of possible DECT-2020 NR applications, but it also limits direct comparison across studies. More consistent reporting of network configurations, traffic characteristics, radio parameters, software and firmware versions, measurement procedures, and statistical summaries would improve reproducibility and enable more meaningful cross-study benchmarking.

G. Cross-Layer Experimental Validation Gaps

The preceding results indicate that the remaining experimental challenges are increasingly cross-layer rather than confined to individual protocol mechanisms. Industrial communication performance emerges from interactions among the PHY, MAC, routing, scheduling, hardware behavior, traffic characteristics, and application requirements. Consequently, successful validation of an individual mechanism does not necessarily demonstrate predictable end-to-end behavior in an integrated system.

A first gap concerns *experimental scale*. Routing, mesh networking, mMTC, and resource-allocation studies frequently target network sizes and traffic conditions that have not yet been reproduced on comparably large physical testbeds. Increasing device density and network depth introduces concurrent channel access, interference, accumulated forwarding delay, synchronization effects, route changes, and contention among heterogeneous traffic flows. Large-scale multi-device experiments are therefore needed to determine whether scalability demonstrated analytically or through simulation persists under physical implementation constraints.

A second gap concerns *cross-layer and application-oriented evaluation*. Industrial applications may simultaneously impose requirements on latency, reliability, throughput, energy consumption, and communication predictability, whereas these properties are often evaluated separately. Experimental campaigns should increasingly combine heterogeneous traffic classes and examine how channel conditions, MAC operation, scheduling, retransmissions, routing, and hardware constraints jointly affect application-visible performance. For latency- and reliability-sensitive communication, distribution tails,

failure events, and temporal variability are particularly important alongside average KPIs.

A third gap concerns *reproducibility and comparability*. Differences in hardware, firmware, network size, topology, traffic generation, radio configuration, propagation environment, and measurement duration make direct cross-study comparison difficult. Reproducible evaluation therefore requires more consistent reporting of experimental configurations, software and firmware versions, traffic characteristics, measurement procedures, and statistical distributions.

Together, these gaps indicate that experimental maturity should not be judged solely by the presence of a hardware implementation. Progress toward practical maturity increasingly requires larger physical systems, cross-layer evaluation, realistic application-oriented traffic, and repeatable integrated-system and industrial experiments. These observations motivate the experimental-maturity framework introduced in Section IV.

IV. EXPERIMENTAL-MATURITY FRAMEWORK

The results in Table I demonstrate that DECT-2020 NR research progresses unevenly from theoretical investigation toward physical implementation and real-world deployment. Research volume alone does not capture this progression: some frequently investigated areas remain dominated by analytical or simulation-based evidence, whereas smaller research areas have already accumulated substantial hardware and practical validation. To characterize these differences systematically, Fig. 3 proposes a five-level experimental-maturity framework.

The framework distinguishes five complementary levels of validation. *Analytical feasibility (L1)* establishes theoretical or model-based evidence that a mechanism, architecture, or performance target is technically plausible. *Simulation validation (L2)* evaluates the proposed mechanism under controlled link-level, system-level, numerical, or event-based conditions and enables investigation of scenarios that may be difficult to reproduce physically. *Prototype validation (L3)* introduces physical implementation through SDR, FPGA, development hardware, or over-the-air experiments, thereby exposing implementation-dependent effects that are typically abstracted in simulation. *Integrated-system validation (L4)* evaluates mechanisms using commercial devices or representative testbeds with real protocol stacks and traffic, providing evidence of their behavior as part of a complete communication system. Finally, *industrial integration (L5)* represents validation through field trials, industrial applications, or long-duration operation under realistic deployment conditions.

These levels should not be interpreted as a mandatory linear development path or as a ranking of research quality. Analytical and simulation-based studies remain essential for theoretical understanding, controlled experimentation, scalability analysis, and exploration of scenarios that may be impractical to reproduce in hardware.

Likewise, individual studies may provide evidence at multiple maturity levels. Rather, progression toward higher levels indicates increasing exposure to implementation-, integration-, and deployment-dependent behavior that cannot be represented completely at lower levels.

The framework is therefore applied to individual technical areas rather than assigning a single maturity level to DECT-2020 NR as a whole. Based on the evidence summarized in Table I, PHY research shows the strongest progression toward Levels 3–5, with substantial prototype, commercial-device/testbed, and field/industrial evidence. Energy research also exhibits comparatively strong hardware maturity despite its smaller publication base. In contrast, MAC and routing/mesh research are extensively studied but remain predominantly supported by analytical and simulation-based evaluation. Scheduling and resource allocation have only limited higher-level evidence, while URLLC and particularly mMTC remain strongly simulation-oriented.

The framework consequently separates *research activity* from *experimental maturity*. A technical area may contain a large number of studies while still lacking evidence from large-scale physical testbeds, integrated systems, or industrial deployments. Conversely, a smaller research area may exhibit comparatively high experimental maturity when its findings have been reproduced across hardware, commercial-device, and field-level evaluations. This distinction provides a structured basis for identifying where future DECT-2020 NR research can benefit most from progression toward large-scale, reproducible, cross-layer, and application-oriented experimental validation.

V. CONCLUSION

This paper assessed the experimental maturity of DECT-2020 NR through a systematic mapping of 51 primary studies published between 2020 and July 2026. The results reveal a clear progression from analytical and simulation-based investigation toward hardware prototypes, commercial-device testbeds, field trials, and industrial validation. However, simulation remains the dominant evaluation methodology, and evidence from large-scale physical systems, field trials, and industrial measurements remains limited.

Experimental maturity also differs substantially across technical areas. PHY and energy research exhibit comparatively strong hardware and practical validation, whereas MAC, routing/mesh, scheduling and resource allocation, URLLC, and mMTC remain more dependent on simulation or small-scale experimental evidence. The analysis further identifies three cross-cutting challenges: limited experimental scale, insufficient cross-layer and application-oriented evaluation, and heterogeneous experimental methodologies that constrain reproducibility and cross-study comparison.

To characterize these differences, this paper introduced a five-level experimental-maturity framework spanning

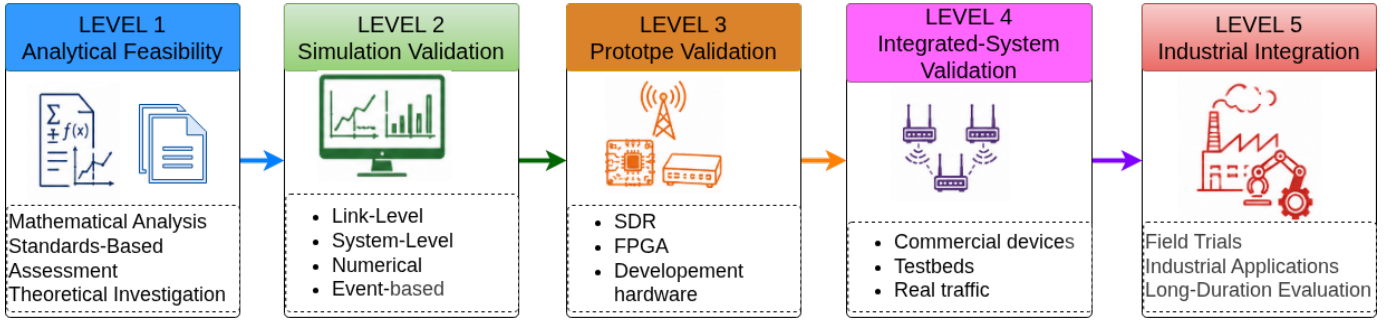


Fig. 3. Proposed five-level experimental-maturity framework for DECT-2020 NR research, progressing from analytical feasibility to industrial integration.

analytical feasibility, simulation validation, prototype validation, integrated-system validation, and industrial integration. The framework distinguishes research activity from experimental maturity and provides a structured basis for assessing how individual technical areas progress toward practical validation.

Future research should prioritize large-scale multi-device and multi-hop testbeds, heterogeneous application-oriented traffic, mobility and interference, long-duration measurements, and reproducible cross-layer benchmarking. Progress in these areas is essential for translating the capabilities demonstrated in analytical, simulation, and small-scale experimental studies into predictable and dependable DECT-2020 NR performance in real-world IoT and industrial deployments.

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